

Contracting Institutions in a Spatial Economy: A Quantitative Analysis of China

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Abstract

Weak contracting institutions cause underinvestment, distort labor demand, and thus shape the spatial distribution of workers and goods within a country. This paper develops a quantitative spatial model with incomplete contracts, in which firms endogenously choose the optimal share of contractible activities subject to contracting costs. We study an institutional reform that uniformly reduces judicial inefficiency by 20% and find an aggregate welfare gain of 5.9%, with substantial heterogeneity across Chinese provinces. Both migration and trade amplify reform-induced welfare gains in initially stronger-institution regions but dampen them in initially weaker-institution regions, with trade exhibiting a stronger amplifying effect and a weaker dampening effect than migration. While migration has little impact on aggregate reform-induced welfare gains, interregional trade amplifies reform-induced aggregate gains by 9% relative to the no-trade scenario.

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1 Introduction

Contractual incompleteness leads to underinvestment and inefficiency (Antras and Helpman, 2004; Acemoglu et al., 2007). While much of the literature studies these effects using cross-country variation in contracting institutions (Nunn, 2007; Levchenko, 2007; Boehm, 2022; Chor and Ma, 2025), substantial variation in contracting institutions also exists within countries such as China (World Bank, 2008) and India (Boehm and Oberfield, 2020). Such institutional distortions depress local labor demand, reduce production efficiency, and shape the spatial allocation of workers and goods.

Internal migration within a country allows workers to relocate away from more distorted regions, an adjustment margin that is largely absent in international settings where workers are typically treated as immobile across countries. Interregional trade provides another adjustment margin by allowing goods produced in regions where production is more efficient to reach consumers elsewhere. Both margins may mitigate the welfare losses caused by pre-reform institutional distortions and further shape the spatial distribution of gains from institutional reform. This paper develops a quantitative spatial model to evaluate how institutional reform affects welfare through its interaction with labor mobility and goods-market integration.

We incorporate region-specific incomplete contracts à la Acemoglu et al. (2007) into the spatial equilibrium framework of Redding (2016) with migration and trade frictions. Consumers in each province derive utility from the consumption of residential land and final goods composed of a continuum of traded varieties. They are mobile across provinces, subject to idiosyncratic preference shocks and migration costs. Firms draw productivity from a Fréchet distribution, produce a variety under perfect competition in each region and trade subject to trade costs, following the framework of Eaton and Kortum (2002).

Unlike Antras and Helpman (2004) and Chor and Ma (2025), we follow Acemoglu et al. (2007) and abstract from global value chains when studying the effect of contracting institutions in a spatial economy.¹ To produce a variety, a firm contracts with contractors to acquire customized tasks. These tasks require local adaptation, monitoring, and coordination, and are produced by contractors using local labor. Firms endogenously choose the share of contractible activities, but doing so involves contracting costs related to legal drafting, verification, judicial support, and

¹Relative to Acemoglu et al. (2007), our model abstracts from technology adoption but introduces spatial labor mobility. We also adapt their framework to a perfectly competitive setting, and endogenize firms' contracting decisions.

enforcement. These costs depend on local governmental contracting institutions. When various contingencies are not well specified in the contract and contracting institutions are weak, fewer activities are made contractible. Hence, contracting institutions govern the share of contractible activities to be carried out by the contractors.

Such relationship-specific customization exposes contractors to be held up by the firm, and contractors underinvest in non-contractible activities relative to the efficient level. This underinvestment reduces their effective supply to firms and lowers overall production efficiency. Meanwhile, the underinvestment will also depress demand for workers in the non-contractible activities for contractors. This translates into exodus of workers to other less-distorted regions. Our quantitative model allows for a closed-form expression linking regional production efficiency to contracting institutions. Hence, better contracting institutions grant a larger share of contractible activities, improve production efficiency and offer higher wages, thereby attracting migration inflows and improves welfare, *ceteris paribus*.

Our quantitative model is calibrated to the Chinese economy, focusing on 30 provinces. We use multiple data sources to estimate province-specific structural parameters, including contracting institutions, amenities, and productivities. We parameterize provincial contracting institutions as a function of two judicial measures, court time and court costs, from the World Bank's *Subnational Doing Business in China 2008*. As contracting institutions directly affect firms' production efficiency and performance, we adopt the nonlinear least squares method to estimate this function by using firm-level sales data from the 2007 Chinese Industrial Enterprise Database. The estimation is robust to a battery of firm-level controls and province-level confounding variables. We use the 2015 One-Percent Population Survey to construct interprovincial migration flows over 2010–2015, which are then used to estimate regional amenities through a gravity equation. Similarly, we use trade flows from the 2010 input-output table to infer regional productivities.

Using our calibrated model, we solve for the baseline spatial equilibrium, and perform several counterfactual analyses to evaluate how institutional reform affects welfare through its interaction with internal migration and interregional trade. In the first exercise, we study an institutional reform that improves judicial efficiency by reducing court time and court costs by 20% in every province. The reform generates an aggregate production efficiency improvement of 10.7% and an aggregate welfare gain of 5.9%, with substantial heterogeneity across provinces. Initially weaker-institution provinces experience larger improvements in production efficiency and greater welfare gains. Meanwhile, both production efficiency improvements and welfare gains decrease with the elasticity of substitution between customized tasks. Intuitively, when

production depends more heavily on inputs that are difficult to substitute for one another, the role of contracting institutions becomes more critical, as they directly affect the supply of non-contractible inputs.

In the second exercise, we shut down migration and re-evaluate the impact of institutional reform on welfare gains. By comparing reform-induced welfare gains in the baseline economy with those in the no-migration economy, we find that migration amplifies reform-induced welfare gains in provinces with initially stronger institutions, such as Guangdong and Jiangsu, but dampens local reform-induced welfare gains in provinces with initially weaker institutions, such as Qinghai and Jilin. Goods-market adjustments explain little of this cross-provincial heterogeneity. Instead, we find that land-market congestion plays an important role. Provinces with initially weaker institutions experience larger improvements in contracting institutions, leading to stronger direct productivity and welfare effects. If migration is allowed, these regions attract migrant inflows from stronger-institution regions after reform. These inflows increase demand for residential land, bid up land rents, and dampen local welfare gains. By contrast, outflows from initially stronger-institution regions alleviate local land-market congestion and reduce land rents, amplifying their welfare gains. Since less-distorted regions experience smaller gains but account for a larger share of the national population, the population-weighted aggregate reform-induced welfare gains differ little between the baseline and no-migration economies.

Finally, we shut down interregional trade and re-evaluate the impact of institutional reform on welfare gains. By comparing reform-induced welfare gains in the baseline economy with those in the no-trade economy, we find that trade also amplifies reform-induced welfare gains in provinces with initially stronger institutions, but dampens local reform-induced welfare gains in provinces with initially weaker institutions. Initially stronger-institution provinces benefit more from trade because interregional trade allows them to access relatively cheaper goods produced in initially weak-institution regions, where institutional reform generates larger efficiency gains. Other channels, including wage and land-rent responses, also matter, but the price-index benefit is the dominant force underlying the heterogeneous pattern of reform-induced welfare gains across provinces: it amplifies welfare gains in stronger-institution regions and dampens them in weaker-institution regions. Interestingly, compared with migration, trade amplifies reform-induced welfare gains more in initially stronger-institution provinces and dampens them less in initially weaker-institution provinces. As a result, trade amplifies aggregate reform-induced welfare gains by 9%.

Beyond the mechanisms above, we provide a decomposition analysis to further shed light

on the heterogeneity of reform-induced welfare gains across provinces. For the migration case, the reform-induced welfare gain in the migration economy relative to the no-migration economy can be decomposed into the difference between the post-reform and pre-reform values of migration. We find that migration mitigates welfare losses from institutional distortions before reform, especially in initially weaker-institution provinces, whose workers can relocate to less distorted regions. This raises the pre-reform value of migration and leaves less remaining welfare loss to be corrected by institutional reform. As a result, migration dampens reform-induced welfare gains in these weak-institution regions. By contrast, pre-reform population inflows into less-distorted regions increase land congestion and reduce the pre-reform value of migration there. Hence, migration is more likely to amplify reform-induced welfare gains in strong-institution regions. A similar logic applies to the trade scenario.

This study contributes to two main strands of literature. First, we contribute to the large literature on resource misallocation. There are various sources underlying resource misallocation, such as firm-level output and labor frictions (Hsieh and Klenow, 2009; Li et al., 2024), financial frictions (Moll, 2014; Midrigan and Xu, 2014), and housing constraints (Hsieh and Moretti, 2019). Instead, we adopt an incomplete-contract approach in the spirit of Acemoglu et al. (2007) with the institutional distortion as a source of resource misallocation, joining related works by Restuccia and Rogerson (2008), Mukoyama and Popov (2020), Boehm and Oberfield (2020), Boehm (2022) and Chor and Ma (2025). Our work is most closely related to Chor and Ma (2025), who develop a model of global sourcing in the presence of contracting frictions and study how these frictions affect country-level welfare and trade patterns. Although we abstract from supply chains and only examine interregional trade within China, we incorporate migration into the model, and study how migration and trade shape the reform-induced welfare gains in a unified framework. Li et al. (2024) study how migration alleviates the negative effect of output and labor distortions at the firm level, but we focus on institutional distortions.

Second, this paper adds to the growing literature on the interaction between internal migration and trade in China. Tombe and Zhu (2019) examine the welfare implications of trade and migration frictions, while Ma and Tang (2020) extend this framework by incorporating firm entry and exit along the extensive margin, and quantitatively study local welfare impacts and spatial spillovers from inter-city migration in China. Zi (2025) finds that liberalizing migration directly increases welfare but reduces the gains from tariff reductions and amplifies their adverse distributional consequences. See also Fan (2019), Hao et al. (2020), and Hsu and Ma (2021), among others. Relative to this literature, we focus on the welfare gains from institutional reform and examine how internal migration and interregional trade shape the aggregate and distributional

effects of these gains across regions.

The remainder of the paper is organized as follows. Section 2 introduces a quantitative spatial equilibrium model that incorporates incomplete contracts. Section 3 details the data and the calibration, while Section 4 presents the main quantitative results. Section 5 concludes.

2 Theoretical Framework

In this section, we model contracting institutions by extending an incomplete-contract model à la Acemoglu et al. (2007) within a quantitative spatial equilibrium framework. We consider an economy consisting of $N = 30$ Chinese provinces (regions), where workers are mobile across regions but face migration frictions. In each region, a firm produces a differentiated variety by combining a continuum of customized tasks. These tasks are specific to the firm's production plant and require local adaptation, monitoring, and coordination. Accordingly, each task is performed by a contractor using local labor. The firm contracts with each contractor, endogenously specifying the range of contractible investments. The varieties are traded in competitive markets subject to iceberg trade costs, as in Eaton and Kortum (2002).

This framework enables us to derive a closed-form expression for regional production efficiency as a function of local contracting institutions. It also facilitates a range of counterfactual analyses that examine the interaction between institutional reform, internal migration and inter-regional trade, as well as their welfare implications.

2.1 Consumer Preferences and Migration Decisions

Worker ω residing in region n derives utility from an idiosyncratic preference shock $a_n(\omega)$, local amenity A_n , consumption of final goods $C_n(\omega)$, and residential land usage $h_n(\omega)$. The utility function is given by:

$$U_n(\omega) = a_n(\omega) A_n \left(\frac{C_n(\omega)}{\beta} \right)^\beta \left(\frac{h_n(\omega)}{1 - \beta} \right)^{1-\beta}, \quad (1)$$

where $0 < \beta < 1$ represents the goods consumption share. The preference shocks $a_n(\omega)$ capture heterogeneity in workers' idiosyncratic preferences for residing in different regions. We assume these shocks are independently and identically distributed (i.i.d.) across both workers and regions,

following a Fréchet distribution:

$$F(a) = e^{-a^{-\kappa}},$$

where the shape parameter κ governs the dispersion of preferences across regions.

The final goods consumption index C_n is defined over a continuum of traded varieties $j \in [0, 1]$:

$$C_n = \left(\int_0^1 c_n(j)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}},$$

where σ denotes the elasticity of substitution between goods. Let $p_{ni}(j)$ denote the price paid by a consumer in region n for one unit of variety j purchased from region i , and $p_n(j)$ is defined as the lowest price of variety j across all regions:

$$p_n(j) \equiv \min_i p_{ni}(j), \forall i \in 1, \dots, N.$$

The corresponding price index P_n for final goods consumption can be expressed as:

$$P_n = \left(\int_0^1 p_n(j)^{1-\sigma} dj \right)^{\frac{1}{1-\sigma}}.$$

Workers maximize their utility subject to the budget constraint:

$$P_n C_n + r_n h_n \leq I_n, \tag{2}$$

where r_n is the land price and I_n represents income per capita in region n . Denote w_n as the wage rate in region n . Land revenue is evenly rebated to local residents. Total income in region n can be written as:

$$I_n L_n = w_n L_n + (1 - \beta) I_n L_n.$$

Thus, the relationship between wage and income can be simplified to:

$$I_n = \frac{1}{\beta} w_n. \tag{3}$$

Maximizing the utility function (1) subject to the budget constraint (2) yields the indirect

utility function:

$$V_n = \frac{a_n(\omega) A_n I_n}{(P_n)^\beta (r_n)^{1-\beta}}. \quad (4)$$

Each worker chooses to reside in the region that offers the highest utility, accounting for both idiosyncratic preferences and migration frictions. Given the Fréchet distribution of preference shocks, the probability that a worker initially registered in region i chooses to migrate to region n is:

$$m_{ni} = \frac{(A_n v_n / \lambda_{ni})^\kappa}{\sum_{k \in N} (A_k v_k / \lambda_{ki})^\kappa}, \quad (5)$$

where $v_n \equiv \frac{I_n}{P_n^\beta r_n^{1-\beta}}$ denotes the real income in region n , and $\lambda_{ni} \geq 1$ captures iceberg-type migration frictions from origin i to destination n . We assume $\lambda_{ii} = 1$, so that workers remaining in their home region do not face any migration cost.

Let $\bar{L}_i$ denote the initial population registered in region i . The equilibrium population in region n is then determined by:

$$L_n = \sum_{i \in N} m_{ni} \bar{L}_i. \quad (6)$$

Accounting for migration frictions, aggregate welfare W is defined as a population-weighted average utility across regions, incorporating the role of idiosyncratic preferences. Specifically, it is given by:

$$W = \sum_n \frac{\bar{L}_n}{\sum_n \bar{L}_n} W_n, \quad (7)$$

where $W_n = A_n v_n (m_{nn})^{-1/\kappa}$ denotes welfare in region n .

2.2 Production and Contracts

Firms Each firm in region n draws an i.i.d. productivity level $z_n(j)$ for producing variety j , following a Fréchet distribution:

$$F(z) = \exp(-T_n z^{-\theta}),$$

where θ governs the dispersion of productivity across varieties, and T_n determines the average productivity level in region n . A firm with productivity $z_n(j)$ produces variety j using an input bundle X_n , which is a CES aggregate of a continuum of customized tasks $x_n(s)$ performed by

contractors indexed by $s \in [0, 1]$. The production function is given by:

$$y_n(j) = z_n(j)X_n,$$

where the input bundle X_n is defined as $X_n \equiv \left(\int_0^1 x_n(s)^\alpha ds \right)^{\frac{1}{\alpha}}$, with $\frac{1}{1-\alpha} > 0$ representing the elasticity of substitution among the tasks $x_n(s)$. Let $\tilde{c}_n$ denote the unit cost of the aggregate input X_n , which will be derived soon. The price paid by a consumer in region n for one unit of variety j produced in region i is:

$$p_{ni}(j) = \frac{d_{ni}\tilde{c}_i}{z_i(j)}$$

where $d_{ni} \geq 1$ represents iceberg trade costs. Specifically, delivering one unit of a variety to region n requires shipping d_{ni} units from region i , with $d_{nn} = 1$ for any region n . The gate price is thus given by $p_{nn}(j)$. The price index of final consumption goods in region n is then expressed as:

$$P_n = \Gamma \left(\frac{\theta - (\sigma - 1)}{\theta} \right)^{\frac{1}{1-\sigma}} \left[\sum_{i \in N} T_i (d_{ni}\tilde{c}_i)^{-\theta} \right]^{-\frac{1}{\theta}}, \quad (8)$$

where $\Gamma(\cdot)$ denotes the Gamma function. The share of goods purchased from region i by consumers in region n is given by:

$$\pi_{ni} = \frac{T_i (d_{ni}\tilde{c}_i)^{-\theta}}{\sum_{k \in N} T_k (d_{nk}\tilde{c}_k)^{-\theta}}. \quad (9)$$

Contractors We assume tasks are specific to the firm's production plant and require local adaptation, monitoring, and coordination. Accordingly, each task $x_n(s)$ is performed by a local contractor s using labor as an input. A task consists of a continuum of investments indexed by $m \in [0, 1]$. We assume, without loss of generality, that activities in the interval $m \in [0, \mu_n]$ are contractible, whereas activities in $m \in (\mu_n, 1]$ are non-contractible. Here, $\mu_n \in (0, 1)$ denotes the equilibrium share of contractible activities in region n . Rather than treating μ_n as exogenous, we allow firms to choose it endogenously by trading off the production-side gains from expanding the contractible set of activities against the associated contracting costs. The details will be specified below.

The customized task is produced according to:

$$x_n(s) = \exp \left(\int_0^{\mu_n} \ln \ell_n^c(m, s) dm + \int_{\mu_n}^1 \ln \ell_n^o(m, s) dm \right),$$

where $\ell_n^c(m, s)$ and $\ell_n^o(m, s)$ denote contractor s 's investments in contractible and non-contractible activities, respectively. When contracts are perfectly complete ($\mu_n = 1$), all investments are contractible, and the task production simplifies to $x_n(s) = \ell_n^c(m, s) \equiv \ell_n^c$, for all $s \in [0, 1]$ and $m \in [0, 1]$. Conversely, under incomplete contracts ($\mu_n < 1$), contractors fulfill the contractible portion of the investments as specified by the firm, while choosing the level of non-contractible investments to maximize their own payoffs. We assume that firms make take-it-or-leave-it offers to contractors. Given the customized nature of the tasks, each contractor's outside option is set to zero. All investments, either contractible or non-contractible, are produced using labor, with unit labor requirements equal to one.

SSPE with Contractual Incompleteness We extend the approach of [Acemoglu et al. \(2007\)](#) by endogenizing firms' choice of the share of contractible activities. Conditional on this extension, we analyze a *symmetric subgame perfect equilibrium* (SSPE) between the firm and its contractors, and adopt the Shapley value as the solution concept for the resulting multilateral bargaining game. The sequence of the game between the firm and a contractor in region n proceeds as follows:

- 1 **Choice of Contractible Share:** The firm first chooses the optimal contractible share $\mu_n \in (0, 1)$, which determines the contractible range of activities in a contract.
- 2 **Contract Offer:** The firm offers a contract $[\{\ell_n^c(m, s)\}_{m=0}^{\mu_n}, t_n]$ to each contractor s . Here, $\ell_n^c(m, s)$ specifies the contractible investment level for contractor s , and t_n represents an upfront payment to contractor s by the firm, which may be positive or negative.
- 3 **Contract Acceptance:** Potential contractors observe the contractible share μ_n and decide whether to accept the offered contracts. Upon acceptance, contractors hire workers to perform the tasks.
- 4 **Investment Decisions:** For contractible investments $m \in [0, \mu_n]$, the contractor s invests exactly $\ell_n^c(m, s)$ as specified in the contract. For non-contractible investments $m \in (\mu_n, 1]$, contractors determine their optimal investment levels accordingly, anticipating the ex post distribution of total revenue.

- 5 **Revenue Bargaining:** The firm and contractors negotiate the division of total revenue, with contractors potentially withholding their specific non-contractible investments as a bargaining chip during the negotiation.
- 6 **Production and Revenue Distribution:** Output is produced and sold, and the resulting revenue is distributed according to the bargaining agreement.

The SSPE is characterized by backward induction. For a contractor s in region n , the contractible investment is precisely fulfilled as specified in the contract. For the non-contractible investments, let $\ell_n^o(-s)$ denote the investment chosen by all contractors except s . Given these investments, the contractors and the firm engage in multilateral Shapley bargaining. The Shapley value for contractor s in this context is denoted by $SV_n(s)$, and an explicit formula for it is provided in the bargaining section below.

Bargaining We now derive the Shapley values in this game, following the framework established by [Shapley \(1953\)](#) and [Osborne and Rubinstein \(1994\)](#). In a bargaining game with a finite number of players, each player's Shapley value is the average of their marginal contributions to all coalitions that consist of players ordered below them in all feasible permutations. To compute the Shapley value for each contractor s , we begin by noting that the firm is an essential player in this bargaining game.² When a coalition includes the firm and a measure ι of contractors, the marginal contribution of contractor s in region n is given by

$$m_n(s, \iota) = \frac{\partial \bar{R}_n(\iota)}{\partial \iota},$$

where the revenue $\bar{R}_n^\iota$ is expressed as:

$$\bar{R}_n(\iota) = p_{nn}(j)z_n(j) \left(\int_0^\iota \left(\exp \left(\int_0^{\mu_n} \ln \ell_n^c(m, k) dm + \int_{\mu_n}^1 \ln \ell_n^o(m, k) dm \right) \right)^\alpha dk \right)^{\frac{1}{\alpha}},$$

with the last input ι being provided by contractor s . Consequently, the marginal contribution of the contractor s to firm $z_n(j)$ is derived as:

$$m_n(s, \iota) = \frac{1}{\alpha} \iota^{\frac{1}{\alpha}-1} p_{nn}(j)z_n(j) \left(\frac{\ell_n^o(s)}{\ell_n^o(-s)} \right)^{\alpha(1-\mu_n)} (\ell_n^c)^{\mu_n} (\ell_n^o(-s))^{1-\mu_n}. \quad (10)$$

²If a coalition does not include the firm, its output is zero, regardless of the coalition's size. Consequently, the marginal contribution of contractor s is zero when a coalition excludes the firm. As we have a continuum of contractors, we follow [Acemoglu et al. \(2007\)](#) to divide the unit measure of contractors equally into Λ segments, each of which represents a contractor. We then obtain the limit expression when Λ approaches infinity.

The firm is included in the production coalition with probability $\frac{\iota}{1}$ when there are ι contractors in total. The Shapley value of a contractor $s \in [0, 1]$ is defined as the average marginal contribution of the contractor to the revenue:

$$\begin{aligned} SV_n(s) &= \int_0^1 \frac{\iota}{1} m_n(s, \iota) d\iota \\ &= \frac{1}{1+\alpha} p_{nn}(j) z_n(j) \left(\frac{\ell_n^o(s)}{\ell_n^o(-s)} \right)^{\alpha(1-\mu_n)} (\ell_n^c)^{\mu_n} (\ell_n^o(-s))^{1-\mu_n}. \end{aligned}$$

Thus, each contractor $s \in [0, 1]$ receives an equal share of $\frac{1}{1+\alpha}$ of the total revenue of any firm $z_n(j)$. Since we focus on the SSPE, each contractor will choose the same optimal level of non-contractible investments, i.e., $\ell_n^o(s) = \ell_n^o(-s) \equiv \ell_n^o$. The output $y_n(j)$ of firm $z_n(j)$ simplifies to:

$$y_n(j) = z_n(j) (\ell_n^c)^{\mu_n} (\ell_n^o)^{1-\mu_n}. \quad (11)$$

Therefore, the Shapley value for any contractor s is given by:

$$SV_n(s) = \frac{1}{1+\alpha} p_{nn}(j) y_n(j), \text{ for all } s \in [0, 1]. \quad (12)$$

Denote the remaining share $\gamma = \frac{\alpha}{1+\alpha}$ to be the fraction of the revenue accruing to the firm. Intuitively, a larger α implies greater substitutability among customized tasks, which weakens the bargaining power of contractors. Each contractor in region n , taking as given the upfront transfer t_n , the contractible investment ℓ_n^c , and the non-contractible investment of other contractors $\ell_n^o(-s)$ chooses its own non-contractible investment level to maximize:

$$\max_{\ell_n^o} SV_n(s) + t_n - \mu_n \ell_n^c w_n - (1 - \mu_n) \ell_n^o w_n, \quad (13)$$

subject to the Shapley value equation (12). Under symmetry, $\ell_n^o(s) = \ell_n^o(-s)$, so the optimal choice of non-contractible investment is:

$$\ell_n^o = \left(\gamma \frac{p_{nn}(j) z_n(j)}{w_n} \right)^{\frac{1}{\mu_n}} \ell_n^c. \quad (14)$$

Optimal Contractible Share We proceed to solve the firm's problem of choosing the optimal contractible share. We assume that the contracting cost takes the form $\frac{\mu_n \ell_n^c w_n}{\xi_n}$, which increases

with the wage bill associated with contractible activities, $\mu_n \ell_n^c w_n$, and decreases with local contracting institutions, ξ_n . The intuition behind this specification is that expanding the contractible set of activities requires additional legal drafting, verification, documentation, monitoring, and judicial support to specify and enforce a larger number of contingencies. Better contracting institutions reduce the costs associated with these activities. Under this formulation, firms face a simple trade-off: expanding the contractible share helps mitigate hold-up distortions, but it also requires greater contracting costs, especially in regions where the government is less efficient at enforcing contracts. This specification preserves the tractability of the model while providing a natural link between firms' contracting choices and the local contracting institutions. We treat the local contracting institutions ξ_n as exogenous to firms and infer ξ_n from the data.

By backward induction, given the contractible share, the firm maximizes its profit by deciding the optimal contractible investment:

$$\max_{\ell_n^c} \gamma p_{nn}(j) y_n(j) - t_n - \frac{\mu_n \ell_n^c w_n}{\xi_n},$$

subject to the contractors' participation constraint:

$$t_n + \frac{1}{1+\alpha} p_{nn}(j) y_n(j) \geq (\mu_n \ell_n^c + (1 - \mu_n) \ell_n^o) w_n.$$

The firm extracts all the surplus from its contractors so that the participation constraint is binding. Substituting this constraint into the firm's problem yields:

$$\max_{\ell_n^c} p_{nn}(j) y_n(j) - \left(\left(1 + \frac{1}{\xi_n}\right) \mu_n \ell_n^c + (1 - \mu_n) \ell_n^o \right) w_n.$$

Using the production function (11) and the relationship between contractible and non-contractible investments (14), both ℓ_n^c and ℓ_n^o can be rewritten as functions of $y_n(j)$. Consequently, the total cost incurred by a firm with productivity $z_n(j)$ is:

$$\begin{aligned} TC_n &= \left[\left(1 + \frac{1}{\xi_n}\right) \mu_n \ell_n^c + (1 - \mu_n) \ell_n^o \right] w_n \\ &= \gamma \left(\frac{\left(1 + \frac{1}{\xi_n}\right) \mu_n}{\left(\gamma \frac{p_{nn}(j) z_n(j)}{w_n} \right)^{\frac{1}{\mu_n}}} + (1 - \mu_n) \right) p_{nn}(j) y_n(j). \end{aligned}$$

Under perfect competition, the price of each variety equals its marginal cost:

$$p_{nn}(j) = \frac{\partial TC_n}{\partial y_n(j)} = \frac{B(\mu_n, \gamma, \xi_n) w_n}{z_n(j)}, \quad (15)$$

where

$$B(\mu_n, \gamma, \xi_n) = \frac{1}{\gamma} \left(\frac{\left(1 + \frac{1}{\xi_n}\right) \mu_n \gamma}{1 - (1 - \mu_n) \gamma} \right)^{\mu_n}. \quad (16)$$

The unit cost of the input bundle X_n is thus given by

$$\tilde{c}_n = B(\mu_n, \gamma, \xi_n) w_n. \quad (17)$$

In the last step of solving the game, the firm chooses its optimal contractible share to minimize its unit cost. This is equivalent to minimizing $B(\mu_n, \gamma, \xi_n)$ by choosing μ_n , taking the wage rate w_n as given. We first study the properties of $B(\mu_n, \gamma, \xi_n)$ in Lemma 1, and then the properties of the optimal contractible share in Proposition 1. We relegate the proofs to Appendix A.

Lemma 1. When $\mu_n \in (0, 1)$, $\gamma \in (0, 1)$, $\xi_n > 0$, the following properties hold:

- (1) $B(\mu_n, \gamma, \xi_n) \geq 1$ for all admissible parameter values;
- (2) $B(\mu_n, \gamma, \xi_n)$ is strictly decreasing in γ ;
- (3) $B(\mu_n, \gamma, \xi_n)$ is strictly decreasing in ξ_n ;
- (4) $B(\mu_n, \gamma, \xi_n)$ is strictly convex in μ_n .

Lemma 1 establishes that $B(\mu_n, \gamma, \xi_n)$ behaves as a well-defined cost wedge in the model. The cost wedge disappears ($B(\mu_n, \gamma, \xi_n) = 1$) only when $\mu_n = 1$ and $\xi_n = \infty$. Smaller γ means a greater complementarity between customized tasks. Such weaker competition among task contractors induces them to offer lower non-contractible investments, results in greater distortion, widens the gap between ℓ_n^c and ℓ_n^o , and thus leads to larger $B(\mu_n, \gamma, \xi_n)$. In addition, stronger contracting institutions ξ_n lower unit costs. The strict convexity of B in μ_n is crucial for the firm's cost minimization problem, as it guarantees that the choice of the contractible share is well-behaved. We denote the optimal contractible share chosen by the firm as μ_n^* such that

$$\mu_n^* \equiv \arg \min_{\mu_n \in [0,1]} B(\mu_n, \gamma, \xi_n), \quad (18)$$

with $\gamma \in (0, 1)$ and $\xi_n > 0$. The properties of μ_n^* are summarized in Proposition 1.

Proposition 1.

- (1) The optimal contractible share μ_n^* exists and is unique;
- (2) Define $\bar{\xi} = \frac{1}{\exp(\gamma-1)/\gamma-1}$. If $\xi_n < \bar{\xi}$, then the optimum is interior, $\mu_n^* \in (0, 1)$, and is characterized by the first-order condition. Otherwise, the optimum is attained at the boundary, $\mu_n^* = 1$;
- (3) The optimal contractible share μ_n^* is weakly increasing in ξ_n , and strictly increasing whenever the optimum is interior.

The monotonic relationship between the optimal contractible share μ_n^* and the local contracting institutions ξ_n is intuitive. It indicates that efficient local contract enforcement alleviates firms' contracting burden, thereby motivating firms to design more contractible activities with their contractors. Interestingly, note that all the firms in the same region choose the same optimal contractible share. The term $B(\mu_n^*, \gamma, \xi_n)^{-1}$ then represents a measure of *derived efficiency* and inversely captures the distortions arising from incomplete contracts and firms' contracting costs for region n .

2.3 General Equilibrium

Total expenditure for goods in region n , denoted by E_n , is given by:

$$E_n = \beta I_n L_n.$$

Using the established relationship between wage and income in equation (3), the goods market clearing condition implies:

$$w_i L_i = \sum_{n=1}^N \pi_{ni} w_n L_n. \quad (19)$$

Land market clearing implies that the equilibrium land rent can be determined from the equality of land income and expenditure:

$$r_n = \frac{(1 - \beta)I_n L_n}{\bar{H}_n} = \frac{1 - \beta}{\beta} \frac{w_n L_n}{\bar{H}_n}, \quad (20)$$

where $\bar{H}_n$ is the fixed land endowment in region n .

Given local contracting institutions $\{\xi_n\}$, land endowments $\{\bar{H}_n\}$, amenities $\{A_n\}$ and productivities $\{T_n\}$, along with migration and trade frictions $\{\lambda_{ni}, d_{ni}\}$, the general equilibrium with incomplete contracts can be represented by the migration share $\{m_{ni}\}$, the measure of workers $\{L_n\}$ in each region $n \in N$, the trade share $\{\pi_{ni}\}$ and the contractible share μ_n^* such that equations (5), (6), (9) and (18) hold, with the associated price system $\{w_n, I_n, P_n, \tilde{c}_n, r_n\}$ such that equations (3), (8), (17), (19) and (20) hold.³

3 Parameterization

We calibrate the model to the Chinese economy using data from 30 provinces. The productivity dispersion parameter is set to $\theta = 4$, following [Simonovska and Waugh \(2014\)](#) and [Tombe and Zhu \(2019\)](#). We set the income elasticity of migration to $\kappa = 1.5$, within the range of 1.2–1.8 reported for China in [Tombe and Zhu \(2019\)](#). Consistent with [Ahlfeldt et al. \(2015\)](#) and [Tan et al. \(2020\)](#), we set the land share $1 - \beta = 0.1$.⁴ Our treatment of material inputs as a composite intermediate input follows [Peter et al. \(2023\)](#), who estimate a plant-level elasticity of substitution across material input categories of 2.5 using an instrumental variables approach. This implies $\alpha = \frac{2.5-1}{2.5} = 0.6$.⁵

3.1 Judicial Costs, Contracting Institutions and Optimal Contractible Share

We link judicial cost data to the contracting institutions ξ_n , which is then used to back out the firms' optimal contractible share μ_n^* in region n . We use two measures of judicial inefficiency for the capital cities of 30 provinces from the World Bank's *Subnational Doing Business in China*

³Note that the equilibrium population in each region can be further divided into workers engaged in contractible and non-contractible investment activities, ℓ_n^c and ℓ_n^o , respectively. Using equation (14) and firms' zero profit condition, we have $\frac{\ell_n^o}{\ell_n^c} = (1 + \frac{1}{\xi_n}) \frac{\gamma \mu_n}{1 - \gamma(1 - \mu_n)}$, $\forall z_n(j)$.

⁴[Ahlfeldt et al. \(2015\)](#) estimate the housing expenditure share to be 0.25, while [Tan et al. \(2020\)](#) calculate the land share of housing value in China to be 0.39. The product of these two values yields $1 - \beta = 0.0975 \approx 0.1$.

⁵We will experiment with alternative values of α in robustness checks, as shown in Appendix B.1.

2008 dataset. The first is Court Time, defined as the number of calendar days required to enforce a standard commercial contract, from filing to payment. The second is Court Costs, defined as total official fees, including court, attorney, and enforcement charges, expressed as a percentage of the claim value, where the claim value is set at twice the provincial GDP per capita. Due to data limitations, each provincial capital's data is assigned to the entire province. Let $\widetilde{cost}_n$ and $\widetilde{time}_n$ denote the standardized values of these two measures, respectively, using their cross-provincial means and standard deviations.

We assume that local contracting institutions follow a logistic functional form:

$$\xi_n(\mathbf{a}) = \bar{\xi} \frac{\exp(h_n(\mathbf{a}))}{1 + \exp(h_n(\mathbf{a}))}, \quad \bar{\xi} = \left(\frac{\exp(\gamma - 1)}{\gamma} - 1 \right)^{-1},$$

with

$$h_n(\mathbf{a}) = a_0 + a_1 \widetilde{cost}_n + a_2 \widetilde{time}_n,$$

where $\mathbf{a} = (a_0, a_1, a_2)$ is the parameter vector to be estimated. This transformation guarantees $0 < \xi_n(\mathbf{a}) < \bar{\xi}$ and therefore ensures that the implied optimal contractible share lies in the interior of the unit interval. Given ξ_n , the firm's optimization problem determines the optimal contractible share μ_n^* , which in turn determines the derived efficiency term $B(\mu_n^*, \gamma, \xi_n)^{-1}$.

We infer the effect of local contracting institutions on firms' sales following [Akerman et al. \(2013\)](#) in estimating the Mincerian wage regression. We jointly estimate the parameter vector $\Theta \equiv (\mathbf{a}', \beta_0, \beta_1, \boldsymbol{\delta}')$ by nonlinear least squares (NLS) using firm-level production data from the 2007 Chinese Industrial Enterprise Database. The estimating equation is

$$\ln y_{fn} = \beta_0 + \beta_1 \ln B(\mu_n^*, \gamma, \xi_n)^{-1} + \mathbf{X}'_{fn} \boldsymbol{\delta} + \eta_{j(f)} + \varepsilon_f, \quad (21)$$

where $\ln y_{fn}$ is log sales of firm f in province n ; $\mathbf{X}_{fn}$ are control variables, $\eta_{j(f)}$ denotes two-digit industry fixed effects, and ε_f is the idiosyncratic error.

By minimizing the sample sum of squared residuals implied by equation (21), we have

$$\hat{\Theta} = \arg \min_{\Theta} \sum_{f,n} (\ln y_{fn} - \beta_0 - \beta_1 B(\mu_n^*, \gamma, \xi_n)^{-1} - \mathbf{X}'_{fn} \boldsymbol{\delta} - \eta_{j(f)})^2.$$

Table 1 reports the NLS estimates under two specifications. Column 1 controls for two-digit industry fixed effects and firm-level inputs, including log capital, log labor, and log materials. To

Table 1: Firm-level estimation results

	Spec 1	Spec 2
a_0	-0.250^{**} (0.099)	-0.244^{**} (0.108)
a_1 : cost	-0.517^{***} (0.037)	-0.498^{***} (0.039)
a_2 : time	-0.870^{***} (0.066)	-0.815^{***} (0.068)
$\beta_1 : -\ln B_n$	1.324^{***} (0.100)	1.368^{***} (0.116)
Observations	291,436	291,436
R^2	0.558	0.559

Notes: Both specifications include two-digit industry fixed effects. Spec 1 controls for firm-level variables (log capital, log labor and log materials) and province-level variables (GDP per capita, human capital, and FDI). Spec 2 additionally controls for firm-level variables, including log firm age, exporter status and firm ownership. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

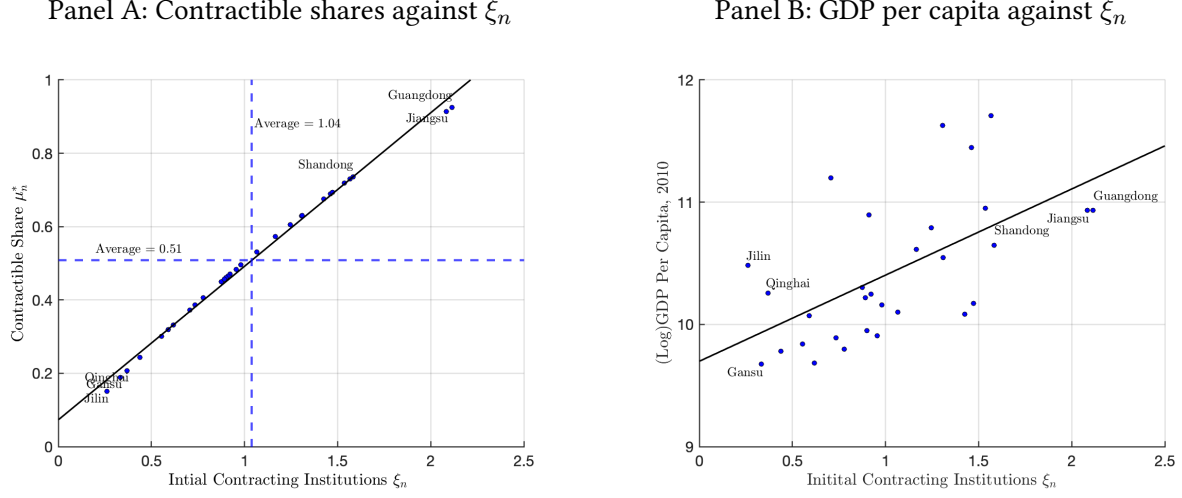
isolate the role of provincial contracting institutions, it also controls for potential province-level confounders, including GDP per capita, human capital, and FDI. Column 2 further adds firm-level controls, including log firm age, exporter status, and firm ownership.

The coefficients on court costs and court time are negative and statistically significant, indicating that higher procedural costs and longer judicial delays are associated with weaker contracting institutions. We use the specification in Column 2 as the baseline. The estimated coefficient on the derived efficiency term, $-\ln B_n$, is positive and statistically significant, with $\hat{\beta}_1 = 1.368$. These estimates are consistent with the mechanism of the model: provinces with stronger contracting institutions support a higher contractible share, which improves production efficiency and is associated with higher firm sales, conditional on observed inputs and controls.

Using the baseline estimates, we recover province-level contracting institutions, ξ_n , and the optimal contractible share, μ_n^* .⁶ Panel A of Figure 1 plots their positive relationship. Guangdong and Jiangsu exhibit the highest optimal contractible shares, exceeding 0.9, whereas Jilin and Gansu have the lowest, below 0.2; the cross-province average is 0.51. Panel B plots log GDP per

⁶The World Bank dataset also provides an overall contract enforcement ranking. We find that the model-implied contracting institutions ξ_n are highly correlated with the World Bank's overall contract enforcement ranking (correlation coefficient = -0.96).

Figure 1: Contracting institutions, contractible shares, and development



Notes: Panel A plots the optimal contractible share, μ_n^* , against initial contracting institutions for each province. Panel B plots log GDP per capita against initial contracting institutions.

capita against the contracting institutions, yielding an estimated slope of 0.70 (s.e. 0.19). The positive relationship suggests that richer regions are positively associated with better contractibility.

3.2 Productivities and Amenities

We recover regional productivities from the structural gravity equation implied by the model. Substituting the trade share expression in equation (9) and rearranging yields

$$\ln X_{ni} = \ln \left(T_i \tilde{c}_i^{-\theta} \right) + \ln \left(I_n \Phi_n^{-1} \right) - \theta \ln(d_{ni}),$$

where X_{ni} denotes the trade flow from origin i to destination n , and $\Phi_n = \sum_k T_k (d_{nk} \tilde{c}_k)^{-\theta}$. Thus, the exporter fixed effect absorbs origin-specific productivity and input costs, while the importer fixed effect captures destination demand and price index conditions. We estimate

$$\ln X_{ni} = D_i^{\text{exp}} + D_n^{\text{imp}} - \theta \ln(d_{ni}) + \varepsilon_{ni}$$

via Poisson Pseudo-Maximum Likelihood (PPML) following Santos Silva and Tenreyro (2006). Bilateral trade costs $\{d_{ni}\}$ are constructed from the transportation dataset of Ma and Tang (2024) for 2010, while the trade flow matrix $\{X_{ni}\}$ is sourced from the 2010 provincial input-output table. Let $\hat{D}_i^{\text{exp}}$ denote the estimated exporter fixed effects. Since the input bundle cost is $\tilde{c}_i = w_i B(\mu_i^*, \gamma, \xi_i)$ and we set $w_i = \beta I_i$ as the inferred wage, we recover regional productivity after

accounting for regional input costs and the production efficiency:

$$T_n = e^{\hat{D}_n^{\text{exp}}} \tilde{c}_n^\theta = e^{\hat{D}_n^{\text{exp}}} \times w_n^\theta \times B(\mu_n^*, \gamma, \xi_n)^\theta. \quad (22)$$

Regional amenities are recovered from the migration equation. Taking logarithms of equation (5) yields

$$\ln L_{ni} = \kappa \ln V_n + \ln (L_i \psi_i^{-1}) - \kappa \ln \lambda_{ni},$$

where $\psi_i = \sum_k (V_k / \lambda_{ki})^\kappa$, and λ_{ni} denotes bilateral migration frictions. We estimate this specification via PPML, with bilateral migration flows derived from the 2015 One-Percent Population Survey and bilateral frictions λ_{ni} constructed from the migration cost measures of [Ma and Tang \(2024\)](#).

Let $\hat{D}_n^{\text{des}}$ be the estimated destination fixed effect. Because $\hat{D}_n^{\text{des}} / \kappa = \ln V_n$, and indirect utility is $V_n = A_n I_n / (P_n^\beta r_n^{1-\beta})$, the amenity level in region n is recovered as

$$A_n = \frac{\exp(\hat{D}_n^{\text{des}} / \kappa) P_n^\beta r_n^{1-\beta}}{I_n}.$$

The goods price index P_n is computed from the equilibrium condition in equation (8). The residential land price r_n is defined as the average transaction price per unit area of residential land in province n during 2008–2018, using parcel-level transaction records from the Ministry of Natural Resources land transaction database.⁷

4 Quantitative Analysis

With the calibrated parameters and province-level fundamentals from Section 3, we solve for the baseline spatial equilibrium. We then conduct three counterfactual (CF) exercises to evaluate the general equilibrium effects of institutional reform on productivity and welfare, and to illustrate how migration and trade can amplify or dampen welfare gains arising from institutional reform across provinces.

1. **CF1 – Institutional Reform:** We keep the costs of migration and trade unchanged and uniformly reduce court time and court costs by 20 percent in all provinces. This shock

⁷This dataset provides comprehensive information on every land transaction, including parcel area, total payment, geographic location, transaction year, land-use type, and auction method. We restrict to residential-use parcels and compute the province-level average price per square meter.

improves province-level contracting institutions ξ_n , which in turn affects the optimal contractible share μ_n^* and the derived efficiency term B_n^{-1} . All other exogenous fundamentals are held at their baseline values. This scenario captures the overall effect of institutional reform in the baseline economy.

2. **CF2 — Institutional Reform without Migration:** We shut down labor mobility by setting migration frictions to infinity. Based on this no-migration equilibrium, we implement the same 20 percent reduction in court time and costs as in CF1 and compare the resulting aggregate welfare gains and distributional effects with those under CF1. This comparison isolates the role of migration in shaping reform-induced welfare gains, conditional on the baseline trade-cost structure.
3. **CF3 — Institutional Reform without Trade:** We shut down interprovincial trade by setting trade costs to infinity. Based on this no-trade equilibrium, we implement the same 20 percent reduction in court time and costs as in CF1 and compare the resulting aggregate welfare gains and distributional effects with those under CF1. This comparison isolates the role of trade in shaping reform-induced welfare gains, conditional on the baseline migration-cost structure.

Meanwhile, we denote $x^{\text{CF},\xi}$ as the value of a variable x under counterfactual scenario CF and contracting institutions ξ . We define

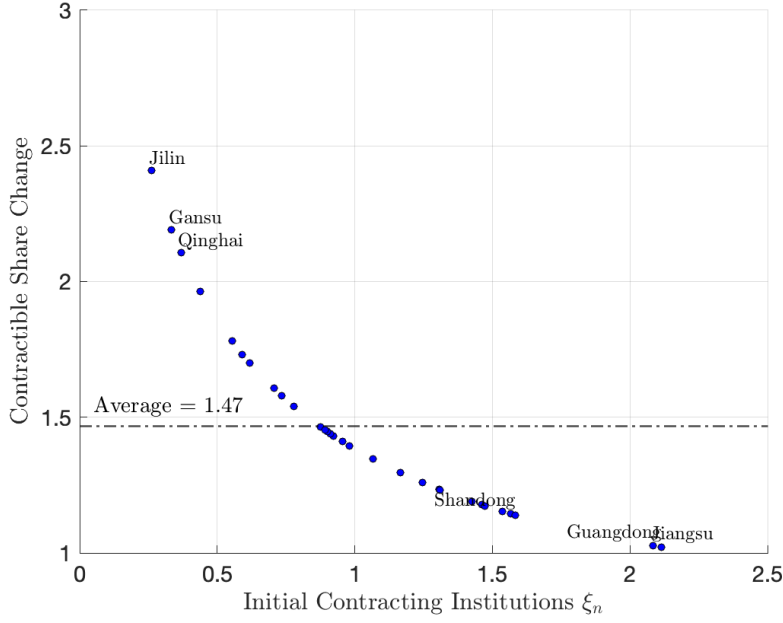
$$\hat{x}^{\text{CF}} \equiv \frac{x^{\text{CF},\xi'}}{x^{\text{CF},\xi}}$$

as the proportional change in x induced by the institutional reform from the initial contracting institutions ξ to ξ' under the same scenario CF.

4.1 CF1: Institutional Reform

Figure 2 plots the post-reform change in the optimal contractible share against initial contracting institutions across provinces. A uniform 20 percent reduction in court time and costs raises the average optimal contractible share by a factor of 1.47. This aggregate effect, however, masks considerable regional heterogeneity. Provinces with initially weaker institutions (higher initial court costs and longer delays) experience larger improvements in contractible shares after the reform. Provinces with initially weaker institutions, such as Jilin and Gansu, see increases exceeding twofold. By contrast, provinces with initially stronger institutions start closer to the

Figure 2: Optimal contractible share change $\widehat{\mu}_n^{*CF1}$



Notes: This figure plots the change in the optimal contractible share $\widehat{\mu}_n^{*CF1}$ under CF1 against initial contracting institutions ξ_n . Provinces with the three highest and lowest contracting institutions are labeled.

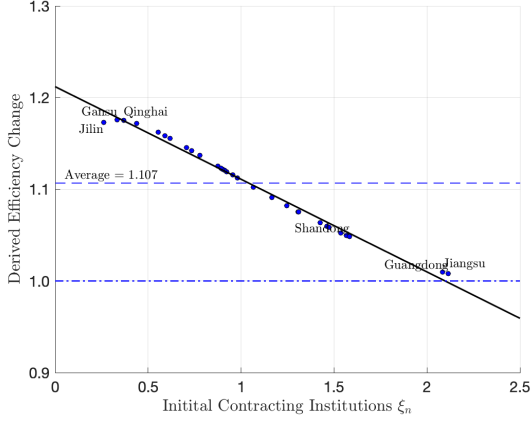
upper bound of feasible contractibility, leaving limited room for further gains. Provinces such as Guangdong and Jiangsu therefore exhibit only marginal improvements in contractibility.

Figure 3 shows the reform's effects on two economically meaningful outcomes: derived efficiency gains (Panel A) and welfare gains (Panel B). Provinces with initially weaker contracting institutions experience larger improvements in both derived efficiency and welfare following the institutional reform. For example, Jilin, Yunnan, and Gansu exhibit increases in derived efficiency and welfare exceeding 15% and 10%, respectively. By contrast, provinces with initially stronger contracting institutions, such as Guangdong and Jiangsu, experience gains of only around 2% in both outcomes. These results are consistent with the intuition that alleviating institutional distortions yields the largest welfare gains in the most distorted regions. The discrepancy between the aggregate welfare gain of 5.9% and the unweighted gain of 8.0% corroborates the small gains but large weights for provinces with initially stronger contracting institutions.

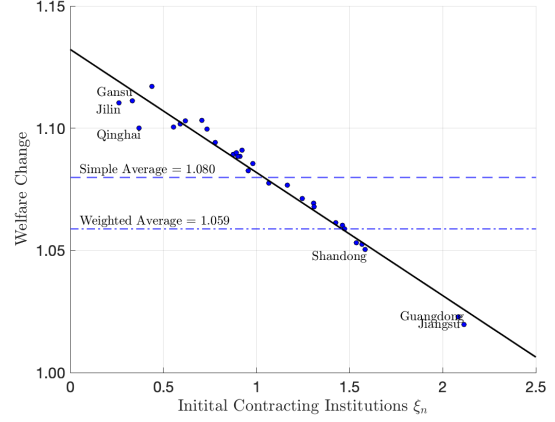
We test the sensitivity of these findings to the elasticity of substitution between customized tasks in Figure B1. When $\alpha = 0.5$ (greater complementarity among tasks), the average production efficiency gain rises to 11.8%, yielding a population-weighted welfare gain of 6.5%. Conversely, when $\alpha = 0.8$ (less complementarity among tasks), the efficiency gain moderates to 8.9%, cor-

Figure 3: Institutional reform, derived efficiency and welfare

Panel A: Derived efficiency change: $\widehat{B}_n^{-1 \text{CF1}}$



Panel B: Welfare change: $\widehat{W}_n^{\text{CF1}}$



Notes: Panel A plots the derived efficiency change under CF1 against initial contracting institutions for each province. Panel B plots the welfare change under CF1 against initial contracting institutions for each province.

responding to a welfare gain of 4.9%. Intuitively, weaker competition among tasks induces task contractors to offer lower non-contractible investments, resulting in greater distortion. Thus, the role of contracting institutions becomes more critical in mitigating misallocation. As a result, improvements in contracting institutions lead to greater gains in derived efficiency and welfare with a smaller α .

4.2 Migration and Trade

We have demonstrated that institutional reform raises welfare by improving local production efficiency. Specifically,

$$\frac{\partial \ln W(\xi, X)}{\partial \xi} > 0,$$

where $W(\xi, X)$ denotes the welfare under contracting institutions ξ and the policy environment X , with X referring to either migration (M) or trade (T).

We next examine whether spatial adjustment margins amplify or dampen the reform-induced welfare gains. Let

$$\widehat{W}_n^{X=1} \equiv \frac{W_n(\xi', X=1)}{W_n(\xi, X=1)}, \quad \widehat{W}_n^{X=0} \equiv \frac{W_n(\xi', X=0)}{W_n(\xi, X=0)},$$

be reform-induced welfare gains under $X = 1$ and $X = 0$, respectively. We then define

$$G_n(X) \equiv \frac{\widehat{W}_n^{X=1}}{\widehat{W}_n^{X=0}}.$$

Thus, $G_n(X)$ measures the reform-induced welfare gains ratio with the policy ($X = 1$) relative to that without the policy ($X = 0$). If $G_n(X) > 1$, the adjustment margin X amplifies reform-induced welfare gains. If $G_n(X) < 1$, the adjustment margin dampens reform-induced welfare gains.

In the counterfactual exercises below, when $X = M$ denotes migration, $X = 1$ corresponds to CF1 and $X = 0$ corresponds to CF2, the institutional reform without migration. When $X = T$ denotes trade, $X = 1$ corresponds to CF1 and $X = 0$ corresponds to CF3, the institutional reform without trade. Therefore,

$$G_n(X) = \begin{cases} \frac{\widehat{W}_n^{\text{CF1}}}{\widehat{W}_n^{\text{CF2}}}, & \text{if } X = M \text{ denotes migration;} \\ \frac{\widehat{W}_n^{\text{CF1}}}{\widehat{W}_n^{\text{CF3}}}, & \text{if } X = T \text{ denotes trade.} \end{cases}$$

We plot $G_n(X)$ against initial contracting institutions ξ_n to examine how the adjustment margin, either migration or trade, affects the reform-induced welfare gains across provinces, and probe the underlying mechanisms.

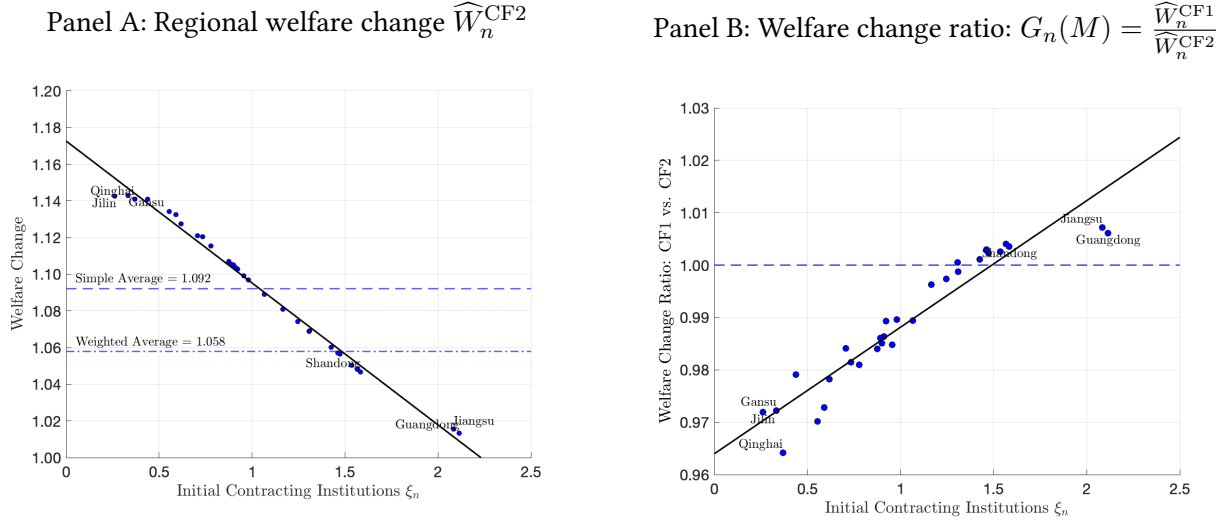
4.2.1 CF2: Institutional Reform without Migration

In this scenario, we completely shut down the interprovincial migration channel. We first compute the equilibrium under the no-migration economy by setting migration costs to infinity. Starting from this equilibrium, we then implement the same institutional reform as in CF1, reducing both court time and court costs by 20 percent, and evaluate the resulting welfare effects under this regime. Recall that $\widehat{x}^{\text{CF2}}$ is the proportional change in a variable x induced by institutional reform when migration is completely shut down.

Panel A of Figure 4 shows that the welfare effects under CF2 exhibit a pattern similar to that in CF1: provinces with initially weaker contracting institutions experience larger welfare gains, whereas provinces with stronger institutions benefit less from institutional reform. The unweighted average provincial welfare gain declines from 9.2% under CF2 to 8.0% under CF1

where migration is allowed. However, the aggregate welfare gain increases from 5.8% under CF2 to 5.9% under CF1. This suggests that migration tends to benefit disproportionately less-distorted provinces, which account for a relatively larger share of the national population.

Figure 4: Welfare changes under institutional reform without migration



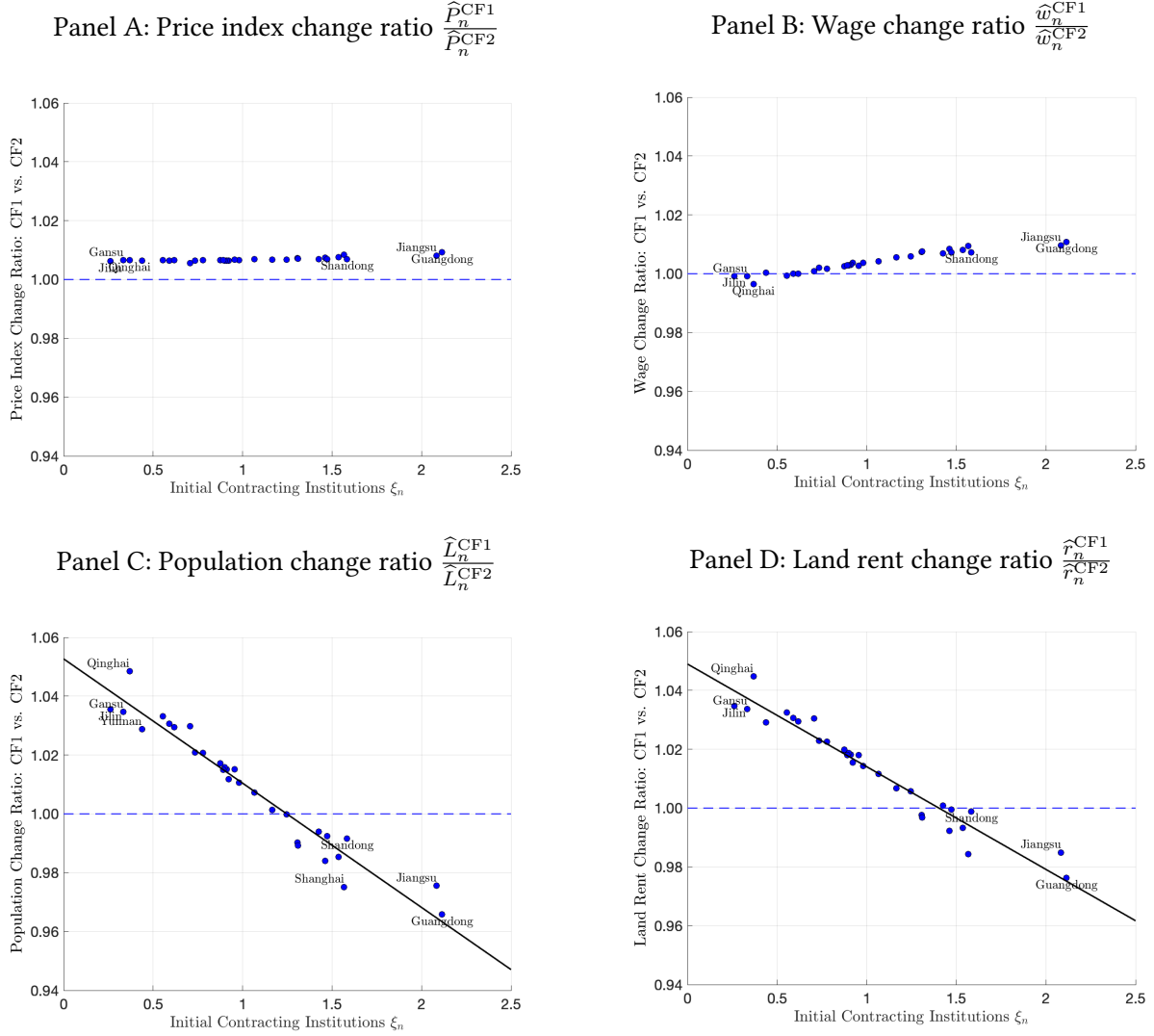
Notes: CF2 considers the same institutional reform as in CF1 but under a no-migration economy. Panel A plots the reform-induced welfare gains under CF2. Panel B plots the reform-induced welfare gains under CF1 relative to that under CF2 across provinces.

We next compare the distribution of welfare gains across provinces under CF1 and CF2. Panel B of Figure 4 plots $G_n(M) = \frac{\widehat{W}_n^{CF1}}{\widehat{W}_n^{CF2}}$ against initial contracting institutions for each province. In provinces with initially weaker contracting institutions, such as Qinghai and Jilin, $G_n(M)$ is below one, indicating that migration dampens local reform-induced welfare gains. By contrast, provinces with initially stronger institutions, such as Guangdong and Jiangsu, have $G_n(M)$ above one, implying that migration amplifies their reform-induced welfare gains.

Figure 5 illustrates the mechanisms underlying these heterogeneous welfare effects. Since trade is still allowed in both CF1 and CF2, firms face similar goods-market access across the two counterfactuals. As a result, price index changes and wage changes induced by institutional reform are comparable under CF1 and CF2, as shown in Panels A and B. This aligns with Redding and Venables (2004), who emphasize that firms' market access to different destinations is a key determinant of the wage rate they can offer.

Although goods-market adjustments explain little of the cross-provincial variation in $G_n(M)$, Panels C and D highlight the role of land-market congestion. Provinces with initially weaker institutions experience larger improvements in contracting institutions, leading to stronger direct

Figure 5: Changes due to institutional reform under CF1 relative to CF2



Notes: CF2 considers the same institutional reform as in CF1 but in an economy without migration. Each panel reports the reform-induced change under CF1 relative to that under CF2.

productivity and welfare effects. Under CF1, these regions attract migrant inflows after reform. These inflows increase demand for residential land, bid up land rents, and dampen local welfare gains. But this channel is shut down under CF2 where migration is prohibited. As a result, both $\frac{\hat{L}_n^{CF1}}{\hat{L}_n^{CF2}}$ and $\frac{\hat{r}_n^{CF1}}{\hat{r}_n^{CF2}}$ are larger in initially weak-institution provinces.

On the other hand, provinces with stronger institutions have limited room for further improvements in contracting institutions and therefore experience smaller direct welfare gains. Under CF1, however, labor mobility allows residents of these stronger-institution regions, such as Guangdong and Jiangsu, to relocate toward regions with initially weaker institutions that expe-

rience larger productivity gains. Such out-migration alleviates local land-market congestion and reduces land rents, amplifying welfare gains relative to the no-migration scenario under CF2.

Interestingly, we can also relate the results above to the change in the *value of migration*. As $G_n(M)$ can be rewritten as

$$G_n(M) = \frac{W_n(\xi', M = 1)/W_n(\xi', M = 0)}{W_n(\xi, M = 1)/W_n(\xi, M = 0)}.$$

Taking logs gives

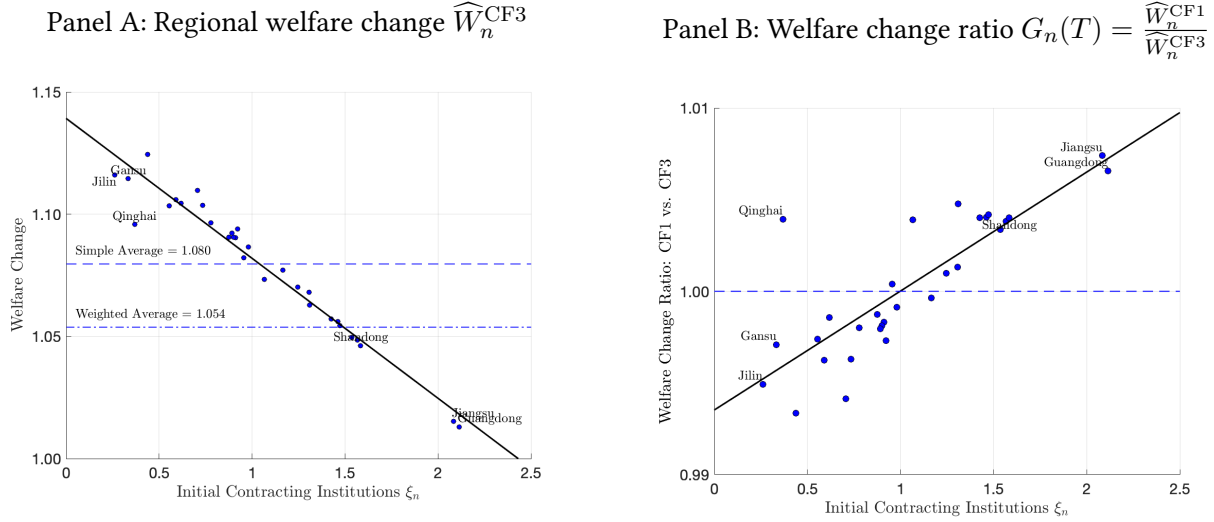
$$\ln G_n(M) = \underbrace{\ln \left[\frac{W_n(\xi', M = 1)}{W_n(\xi', M = 0)} \right]}_{\text{post-reform value of migration}} - \underbrace{\ln \left[\frac{W_n(\xi, M = 1)}{W_n(\xi, M = 0)} \right]}_{\text{pre-reform value of migration}}.$$

This decomposition shows that migration dampens (amplifies) reform-induced welfare gains when its pre-reform value is large (small) relative to its post-reform value.⁸ Panel A of Figure B2 provides evidence that migration can mitigate welfare losses due to institutional distortions before reform, especially in initially weak-institution provinces, whose workers can relocate to less distorted regions. See more details in Appendix B.2. This raises the pre-reform welfare benchmark in the migration economy and leaves less remaining welfare loss to be corrected by institutional reform. In other words, migration will dampen reform-induced welfare gains in these weak-institution regions. On the other hand, pre-reform inflows into less-distorted regions increase land congestion and dampen the pre-reform value of migration in these regions. Hence, migration is more likely to amplify reform-induced welfare gains in strong-institution regions.

Both perspectives, either the role of land-market congestion induced by post-reform in-migration, or workers' pre-reform exodus from more distorted regions, indicate that migration dampens welfare gains from institutional reforms in initially more distorted regions, but amplifies welfare gains in initially less distorted regions. As a result, internal migration plays a critical role in shaping the spatial distribution of welfare gains across regions, despite that the aggregate reform-induced welfare changes little when comparing CF1 and CF2.

⁸Regions with a large pre-reform value of migration could, in principle, have an even larger post-reform value of migration, leading migration to amplify reform-induced welfare gains. Our results, however, show that this is not the case. The pre-reform value of migration tends to be larger (smaller) than the post-reform value of migration in regions with initially weaker (stronger) contracting institutions, leading to $\ln G_n(M) < 0$ ($\ln G_n(M) > 0$), or equivalently $G_n(M) < 1$ ($G_n(M) > 1$), consistent with Panel B of Figure 4.

Figure 6: Welfare change under institutional reform without trade



Notes: CF3 considers the same institutional reform as in CF1 but under a no-trade economy. Panel A plots the reform-induced welfare gains under CF3. Panel B plots the reform-induced welfare gains under CF1 relative to that under CF3 across provinces.

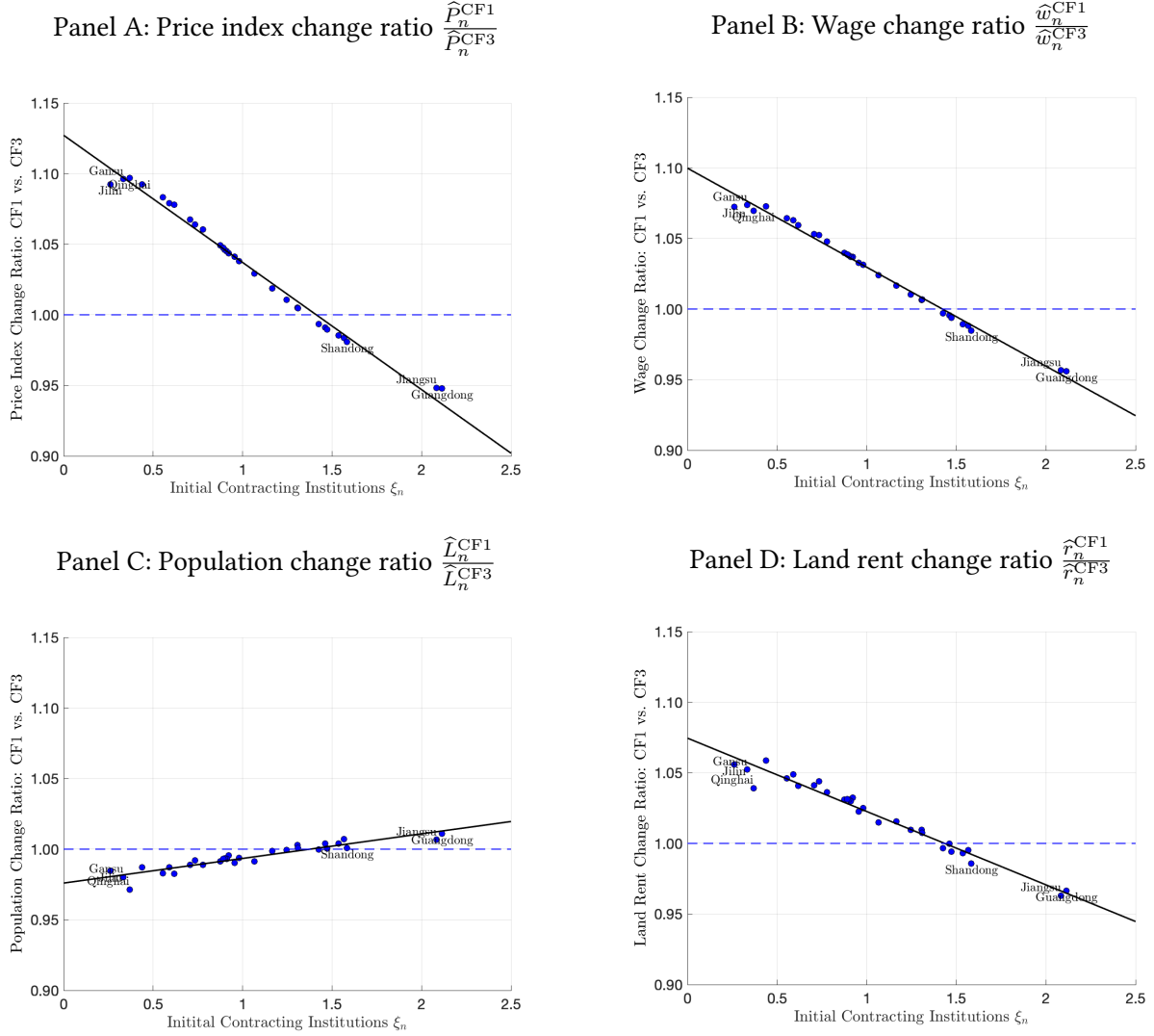
4.2.2 CF3: Institutional Reform without Trade

In this scenario, we completely shut down the interprovincial trade channel. We first compute the equilibrium under the no-trade economy by setting trade costs to infinity. Starting from this equilibrium, we then implement the same institutional reform as in CF1, reducing both court time and court costs by 20 percent, and evaluate the resulting welfare effects under this regime. Recall that $\widehat{x}^{\text{CF3}}$ is the proportional change in a variable x induced by institutional reform when interprovincial trade is completely shut down (CF3).

Panel A of Figure 6 delivers a similar message that provinces with initially weaker contracting institutions experience larger welfare gains, whereas provinces with stronger institutions benefit less from institutional reform when interprovincial trade is shut down. The unweighted average welfare gain under CF3 is comparable to that under CF1. But the aggregate welfare gain under CF3 is 5.4%, lower than 5.9% under CF1. This indicates that trade will amplify the reform-induced welfare gain by $\frac{5.9-5.4}{5.4} = 9\%$. But again, this masks the great heterogeneity of welfare changes across provinces.

We next compare the distribution of welfare gains across provinces under CF1 and CF3. Panel B of Figure 6 plots $G_n(T) = \frac{\widehat{W}_n^{\text{CF1}}}{\widehat{W}_n^{\text{CF3}}}$ against initial contracting institutions for each province. In provinces with initially stronger contracting institutions, such as Guangdong and Jiangsu,

Figure 7: Changes due to institutional reform under CF1 relative to CF3



Notes: CF3 considers the same institutional reform as in CF1 but under a no-trade economy.

$G_n(T)$ is above one, indicating that trade amplifies local reform-induced welfare gains. By contrast, provinces with initially weaker institutions, such as Jilin and Gansu, have values of $G_n(T)$ below one, implying that trade plays a dampening role for these regions.

Figure 7 illustrates the mechanisms underlying these heterogeneous welfare effects. Panels A–D plot the ratios of post-reform changes in the price index, wages, population, and land rents under CF1 relative to those under CF3, respectively.

Under CF1, interprovincial trade allows provinces to access relatively cheaper goods produced in initially weak-institution regions, where institutional reform generates larger efficiency gains. This channel is especially important for initially strong-institution provinces, which expe-

rience relatively small direct efficiency gains but benefit indirectly from relatively cheaper goods produced elsewhere. When trade is shut down under CF3, this price-index benefit is no longer transmitted across provinces, driving up the relative price index under CF3 compared with CF1, especially in initially strong-institution regions. Hence, the price-index change ratio, $\frac{\hat{P}_n^{CF1}}{\hat{P}_n^{CF3}}$, is lower for initially strong-institution regions, as shown in Panel A.

Meanwhile, compared with CF3, interprovincial trade enables initially weak-institution provinces to export the reform-induced productivity gains to the national market. Expanded market access matters more for these initially weak-institution provinces, strengthening local labor demand, thereby offering higher wage rates when compared with strong-institution provinces. On the other hand, weak-institution provinces can experience worker outflows because a lower relative price index makes strong-institution provinces more attractive, as shown in Panel C. The resulting labor outflows shrink local labor supply and further drive up wages, as shown in Panel B. By contrast, initially strong-institution provinces will have lower wage-change ratio because the opposite force operates.

The land-rent response in Panel D reflects the net effect of the population pressure and local income effect, as suggested by the land market clearing condition (20) when local land supply is fixed. In initially weak-institution provinces, though migration outflows tend to depress land rents, the rise in local wages strengthens residential land demand. If the positive income effect dominates the negative population effect, the land-rent change can be larger under CF1 than under CF3. Conversely, in initially strong-institution provinces, land rents may decline more despite population inflows, because lower local wages are capitalized into residential land demand.

To better visualize the relative importance of these forces, we use the same y -axis scale across panels in Figure 7. For initially weak-institution regions, the reform-induced change in the price index dominates the change in wages, contributing to $G_n(T) < 1$. The opposite pattern holds for initially strong-institution regions, resulting in $G_n(T) > 1$.

We can also relate these results to the change in the *value of trade*. We have

$$\ln G_n(T) = \underbrace{\ln \left[\frac{W_n(\xi', T=1)}{W_n(\xi', T=0)} \right]}_{\text{post-reform value of trade}} - \underbrace{\ln \left[\frac{W_n(\xi, T=1)}{W_n(\xi, T=0)} \right]}_{\text{pre-reform value of trade}}.$$

This decomposition shows that trade dampens (amplifies) reform-induced welfare gains

when its pre-reform value is large (small) relative to its post-reform value.⁹ Panel B of Figure B2 provides evidence that trade can mitigate welfare losses due to institutional distortions before reform, especially in initially weak-institution provinces, which can access the markets of less-distorted and more-developed regions. See more details in Appendix B.2. This raises the pre-reform welfare benchmark in the trade economy and leaves less remaining welfare loss to be corrected by institutional reform. In other words, trade will have a stronger dampening effect on reform-induced welfare gains in regions with weaker institutions.

Finally, we combine Panel B of Figure 4 and Panel B of Figure 6 in Figure B4. This comparison illustrates the relative importance of migration and trade in shaping reform-induced welfare gains. Figure B4 shows that $G_n(T)$ lies above $G_n(M)$ for all provinces, and that the two measures converge as contracting institutions become stronger. Compared with migration, trade therefore amplifies reform-induced welfare gains more in initially stronger-institution provinces and dampens them less in initially weaker-institution provinces. This is consistent with our previous observation that migration has little effect on aggregate reform-induced welfare gains, whereas trade amplifies aggregate gains by 9%.

5 Conclusion

In this paper, we demonstrate that contracting institutions shape aggregate and distributional welfare through their interaction with internal migration and interregional trade. Using a quantitative spatial model with incomplete contracts, we show that weaker contracting institutions lead firms to choose a smaller share of contractible activities. This results in underinvestment by contractors in non-contractible activities, distorts labor demand, and influences workers' migration decisions. Our quantitative analysis shows that an institutional reform that uniformly reduces judicial inefficiency by 20% generates an aggregate welfare gain of 5.9%, with substantial heterogeneity across Chinese provinces.

Both internal migration and interregional trade amplify reform-induced welfare gains in initially stronger-institution regions but dampen them in initially weaker-institution regions. Compared with migration, trade has a larger amplifying effect and a smaller dampening effect across

⁹Regions with a large pre-reform value of trade could, in principle, have an even larger post-reform value of trade, leading trade to amplify reform-induced welfare gains. Our results, however, show that this is not the case. The pre-reform value of trade tends to be larger (smaller) than the post-reform value of trade in regions with initially weaker (stronger) contracting institutions, leading to $\ln G_n(T) < 0$ ($\ln G_n(T) > 0$), or equivalently $G_n(T) < 1$ ($G_n(T) > 1$), consistent with Panel B of Figure 6.

provinces. As a result, while migration has little impact on aggregate reform-induced welfare gains, interregional trade amplifies aggregate gains by 9% relative to the no-trade scenario.

Our analysis abstracts from global value chains when examining the implications of contractual incompleteness. Extending the framework to incorporate global production networks and international migration represents a promising direction for future research.

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A Theoretical Appendix

A.1 Proof of Lemma 1

Define

$$x(\mu_n) \equiv \frac{\mu_n \gamma}{1 - (1 - \mu_n) \gamma}.$$

and we have

$$B(\mu_n, \gamma, \xi_n) = \frac{1}{\gamma} \left[\left(1 + \frac{1}{\xi_n} \right) x(\mu_n) \right]^{\mu_n}.$$

(1) Note $\ln B = -\ln \gamma + \mu_n \ln \left(1 + \frac{1}{\xi_n} \right) + \mu_n \ln x(\mu_n)$. Define

$$g(\mu_n) \equiv -\ln \gamma + \mu_n \ln x(\mu_n),$$

and we have

$$g'(\mu_n) = \ln x(\mu_n) + 1 - x(\mu_n).$$

Because $\ln z \leq z - 1$ for all $z > 0$, it follows that

$$\ln x(\mu_n) + 1 - x(\mu_n) \leq 0.$$

Hence $g(\mu_n)$ is decreasing in μ_n . Since $g(1) = -\ln \gamma + \ln \gamma = 0$, we obtain $g(\mu_n) \geq 0$ for all $\mu_n \in (0, 1]$. Therefore

$$\ln B = g(\mu_n) + \mu_n \ln \left(1 + \frac{1}{\xi_n} \right) \geq 0,$$

which implies $B(\mu_n, \gamma, \xi_n) \geq 1$. The inequality is strict unless $\mu_n = 1$ and $\xi_n = \infty$.

(2) Differentiate $\ln B$ with respect to γ :

$$\frac{\partial \ln B}{\partial \gamma} = -\frac{1}{\gamma} + \mu_n \left[\frac{1}{\gamma} + \frac{1 - \mu_n}{1 - (1 - \mu_n) \gamma} \right].$$

Simplifying,

$$\frac{\partial \ln B}{\partial \gamma} = (1 - \mu_n) \left[-\frac{1}{\gamma} + \frac{\mu_n}{1 - (1 - \mu_n) \gamma} \right] = \frac{(1 - \mu_n)(\gamma - 1)}{\gamma [1 - (1 - \mu_n) \gamma]} < 0.$$

Hence $B(\mu_n, \gamma, \xi_n)$ is decreasing in γ .

(3) Differentiate $\ln B$ with respect to ξ_n :

$$\frac{\partial \ln B}{\partial \xi_n} = \mu_n \frac{\partial}{\partial \xi_n} \ln \left(1 + \frac{1}{\xi_n} \right) = -\mu_n \frac{1}{\xi_n^2 \left(1 + \frac{1}{\xi_n} \right)} < 0.$$

Thus $B(\mu_n, \gamma, \xi_n)$ is decreasing in ξ_n .

(4) Differentiate $\ln B$ with respect to μ_n :

$$\frac{\partial \ln B}{\partial \mu_n} = \ln \left[\left(1 + \frac{1}{\xi_n} \right) x(\mu_n) \right] + 1 - x(\mu_n).$$

Since

$$x'(\mu_n) = \frac{\gamma(1-\gamma)}{(1-\gamma+\mu_n\gamma)^2} > 0,$$

we obtain

$$\frac{\partial^2 \ln B}{\partial \mu_n^2} = x'(\mu_n) \left(\frac{1}{x(\mu_n)} - 1 \right) > 0,$$

because $0 < x(\mu_n) < 1$. Therefore

$$\frac{\partial^2 B}{\partial \mu_n^2} = B \left[\left(\frac{\partial \ln B}{\partial \mu_n} \right)^2 + \frac{\partial^2 \ln B}{\partial \mu_n^2} \right] > 0,$$

which proves that $B(\mu_n, \gamma, \xi_n)$ is strictly convex in μ_n .

A.2 Proof of Proposition 1

(1) By Lemma 1, $B(\mu_n, \gamma, \xi_n)$ is strictly convex in μ_n . Hence the minimizer is unique.

(2) Recall from Lemma 1,

$$\frac{\partial \ln B}{\partial \mu_n} = \ln \left[\left(1 + \frac{1}{\xi_n} \right) x(\mu_n) \right] + 1 - x(\mu_n).$$

Therefore,

$$\frac{\partial B}{\partial \mu_n} = B(\mu_n, \gamma, \xi_n) \left\{ \ln \left[\left(1 + \frac{1}{\xi_n} \right) x(\mu_n) \right] + 1 - x(\mu_n) \right\}.$$

First note that

$$\frac{\partial B}{\partial \mu_n} \rightarrow -\infty \quad \text{as } \mu_n \rightarrow 0^+.$$

Thus the minimizer cannot occur at $\mu_n = 0$.

At $\mu_n = 1$, since $x(1) = \gamma$, we obtain

$$\left. \frac{\partial \ln B}{\partial \mu_n} \right|_{\mu_n=1} = \ln \left[\left(1 + \frac{1}{\xi_n} \right) \gamma \right] + 1 - \gamma.$$

Define

$$\bar{\xi} = \frac{1}{\exp(\gamma - 1)/\gamma - 1}.$$

If $\xi_n < \bar{\xi}$, then

$$\ln \left[\left(1 + \frac{1}{\xi_n} \right) \gamma \right] + 1 - \gamma > 0,$$

so $\frac{\partial B}{\partial \mu_n} > 0$ at $\mu_n = 1$. Since $\partial B / \partial \mu_n < 0$ near $\mu_n = 0$ and is strictly increasing in μ_n by convexity, there exists a unique interior minimizer $\mu_n^* \in (0, 1)$.

If instead $\xi_n \geq \bar{\xi}$, then

$$\ln \left[\left(1 + \frac{1}{\xi_n} \right) \gamma \right] + 1 - \gamma \leq 0,$$

so $\frac{\partial B}{\partial \mu_n} \leq 0$ at $\mu_n = 1$. Because $\partial B / \partial \mu_n$ is strictly increasing in μ_n , it follows that $\partial B / \partial \mu_n < 0$ on $(0, 1)$, and therefore B is decreasing on $[0, 1]$. Hence the unique minimizer is $\mu_n^* = 1$.

(3) Consider the interior case. The first-order condition is

$$F(\mu_n, \xi_n) \equiv \ln \left(\frac{\left(1 + \frac{1}{\xi_n} \right) \mu_n \gamma}{1 - (1 - \mu_n) \gamma} \right) + 1 - \frac{\mu_n \gamma}{1 - (1 - \mu_n) \gamma} = 0.$$

By strict convexity, we have

$$F_{\mu_n}(\mu_n, \xi_n) > 0.$$

Moreover,

$$F_{\xi_n}(\mu_n, \xi_n) = \frac{\partial}{\partial \xi_n} \ln \left(1 + \frac{1}{\xi_n} \right) = -\frac{1}{\xi_n^2 \left(1 + \frac{1}{\xi_n} \right)} < 0.$$

Hence, by the implicit function theorem,

$$\frac{\partial \mu_n^*}{\partial \xi_n} = -\frac{F_{\xi_n}}{F_{\mu_n}} > 0.$$

Therefore, μ_n^* is strictly increasing in ξ_n whenever the optimum is interior. Since $\mu_n^* = 1$

for all $\xi_n \geq \bar{\xi}$, it follows that μ_n^* is weakly increasing in ξ_n globally.

B Additional Figures

B.1 Robustness Checks

As a robustness check, Panels A and B of Figure B1 report the CF1 results for $\alpha = 0.5$ and $\alpha = 0.8$, respectively. The corresponding average derived efficiency gains are 11.8% and 8.9%, while the average welfare gains are 6.5% and 4.9%. These results underscore that a lower value of α (greater complementarity among customized tasks) amplifies the derived efficiency and welfare effects of institutional improvements. Intuitively, when production depends more heavily on inputs that are difficult to substitute for one another, the role of contracting institutions becomes more critical. Consequently, improvements in contracting institutions lead to larger gains in efficiency and welfare.

B.2 Pre-reform Values of Migration and Trade

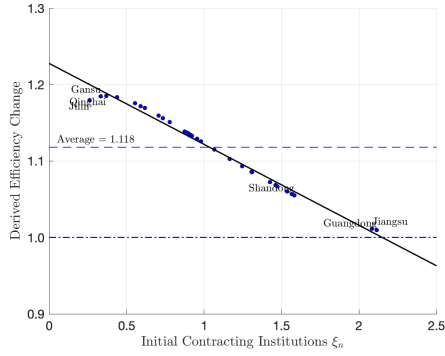
Panel A of Figure B2 shows that the pre-reform value of migration decreases with initial contracting institutions across provinces. Most log values are above zero, indicating that migration improves welfare relative to the no-migration case before reform, especially in provinces with initially weaker contracting institutions. In other words, migration mitigates welfare losses from institutional distortions before institutional reform. However, some provinces still have negative log values of migration. This reflects the general-equilibrium effects of migration. As shown in Figure B3, provinces with initially weaker contracting institutions tend to experience population outflows and lower land rents when migration is allowed, relative to the no-migration case. These adjustments raise welfare in weak-institution regions. The opposite occurs in initially strong-institution regions, such as Guangdong: migration-induced inflows increase land rents and can lower local welfare relative to the no-migration case.

Panel B of Figure B2 shows that the pre-reform value of trade also decreases with initial contracting institutions across provinces. All log values are above zero, indicating that trade improves welfare relative to the no-trade case before reform. Thus, trade also mitigates welfare losses from institutional distortions before reform, especially in provinces with initially weaker contracting institutions.

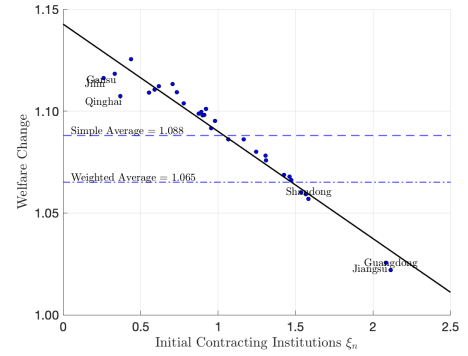
Figure B1: Institutional reform under alternative values of α

Panel A: $\alpha = 0.5$

Panel A: Derived Efficiency Change $\widehat{B}_n^{-1}$

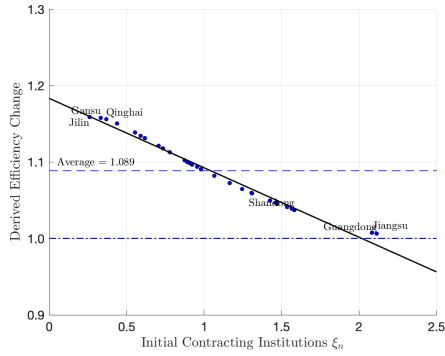


Welfare Change $\widehat{W}_n$

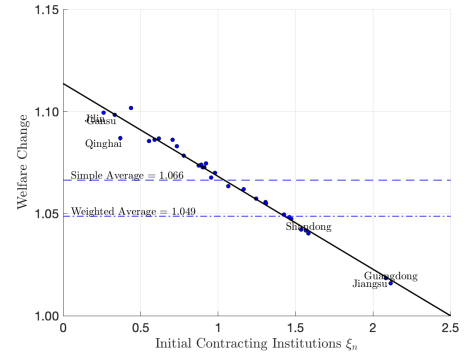


Panel B: $\alpha = 0.8$

Derived Efficiency Change $\widehat{B}_n^{-1}$

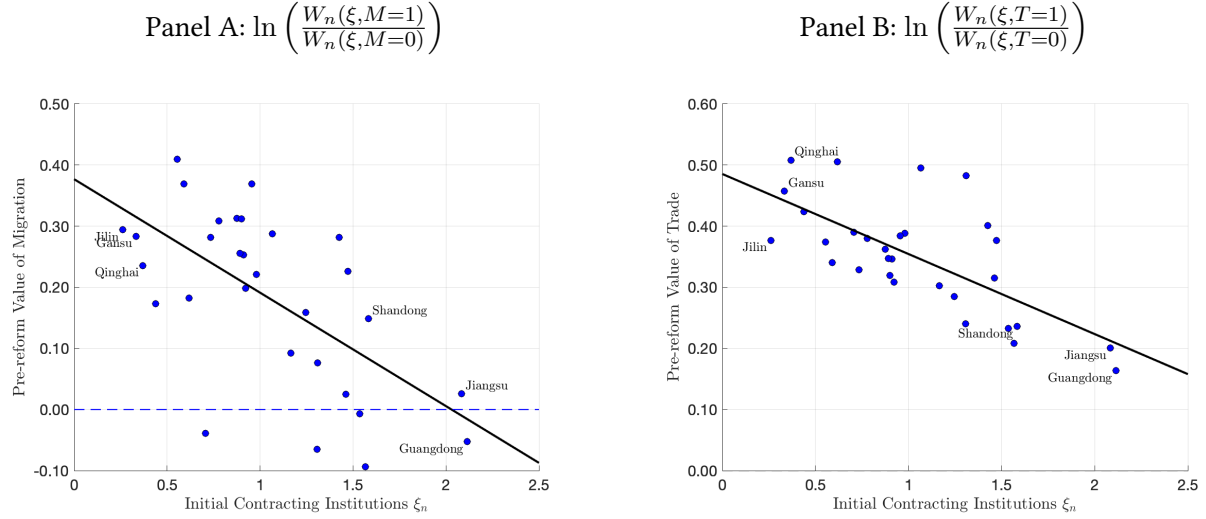


Welfare Change $\widehat{W}_n$



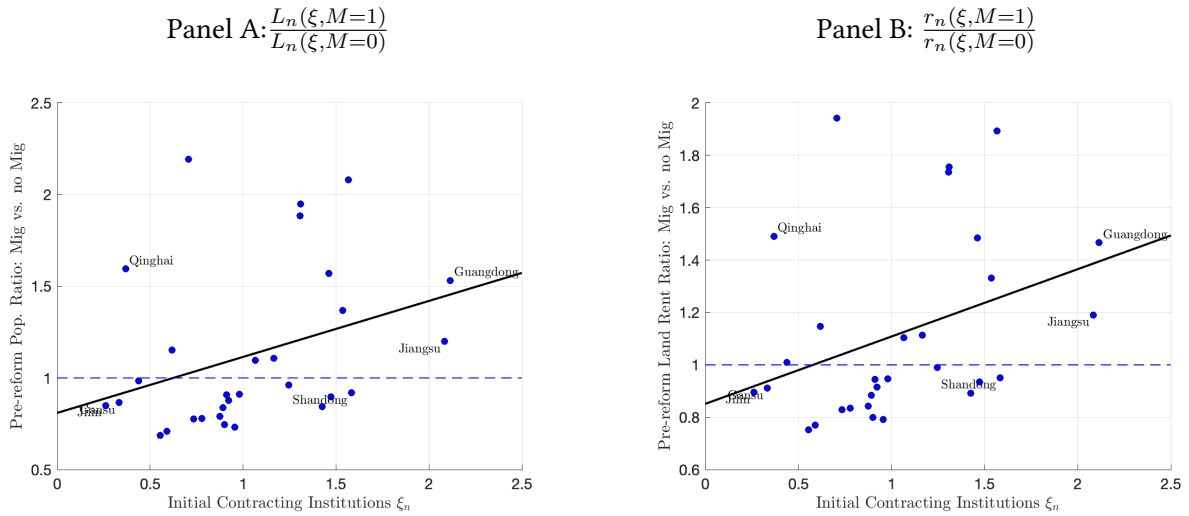
Notes: Panel A reports the results under $\alpha = 0.5$, while Panel B reports the results under $\alpha = 0.8$.

Figure B2: Pre-reform values of migration and trade



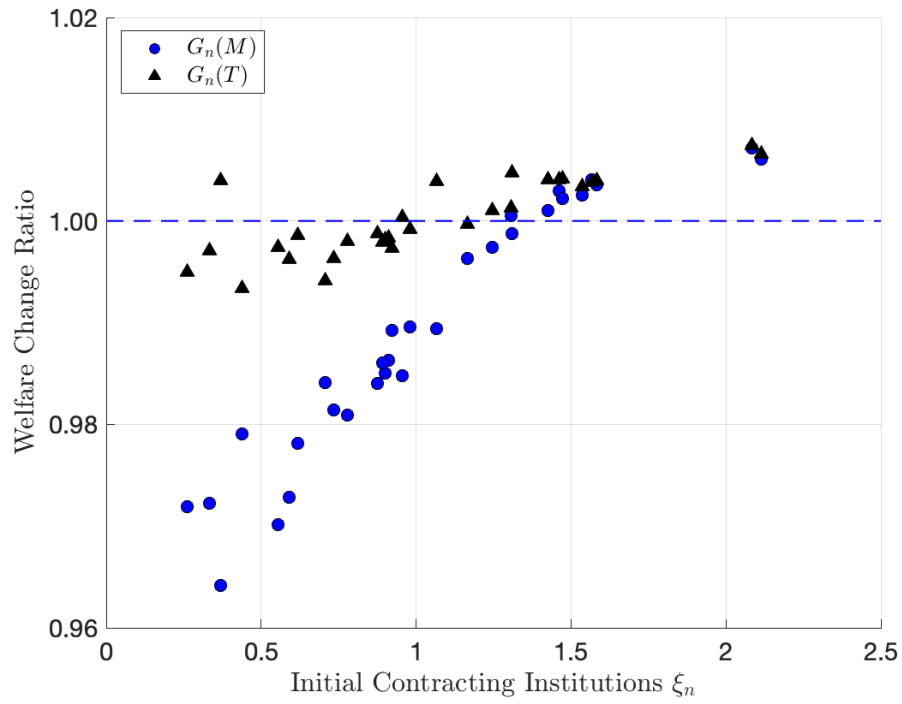
Notes: Panel A plots the pre-reform value of migration against initial contracting institutions ξ_n across provinces. Panel B plots the pre-reform value of trade against initial contracting institutions ξ_n across provinces.

Figure B3: Pre-reform population and land rent ratios: migration vs. no migration



Notes: Panel A plots the pre-reform migration-induced population ratio $\frac{L_n(\xi, M=1)}{L_n(\xi, M=0)}$ against initial contracting institutions ξ across provinces. Panel B plots the migration-induced land rent ratio $\frac{r_n(\xi, M=1)}{r_n(\xi, M=0)}$ against initial contracting institutions ξ across provinces.

Figure B4: The Comparison of $G_n(M)$ and $G_n(T)$



Notes: $G_n(M)$ measures the reform-induced welfare gain under baseline migration relative to the no-migration case. $G_n(T)$ measures the reform-induced welfare gain under baseline trade relative to the no-trade case. We plot both $G_n(M)$ and $G_n(T)$ against initial contracting institutions ξ_n across provinces.